NEO Mathematics — Learner Guide

Accuracy, Powers and Scale · Write and Order Standard Form

Driving Question: *What changes when I adjust one part of a number in standard form?*

What are we exploring?

Some numbers are enormous (the distance to the Sun) and some are tiny (the width of a virus). Standard form is a shared way of writing any of them so they are easy to read, compare and order. In this lesson you will zoom from the micro world to the macro world and discover which part of a number in standard form does the heavy lifting.

Your Stepping Stones

- I can write large and small numbers in standard form.
- I can convert between ordinary numbers and standard form.
- I can order numbers written in standard form.
- I can explain what changing the exponent does compared with changing the coefficient.

Before you begin

- **Useful:** something to write with — the Maths Scratchpad, paper or a whiteboard all work.
- **Optional:** headphones if sound around you is distracting; water; a comfortable place to sit.
- If powers of ten or index laws feel far away, the **Reconnection Routes** button offers a short recap that returns you straight to this lesson.

A possible way through

This is an invitation, not a rule — you can take a different route.

- **Connect (about 5 minutes):** start with "One number, three costumes" and notice what makes standard form special.
- **Explore (about 20 minutes):** open the Micro-to-Macro Scale Explorer. Predict each ordinary number before you check. Then try one change at a time and watch what moves.
- **Create or Apply (about 12 minutes):** draft your Scale Postcard, then try the Practice Companion.
- **Reflect (about 8 minutes):** answer "Which part moves a number furthest?", rest with the calm three-step route, and notice what this lesson nourished.

Working with the Scale Explorer

- Choose a coefficient (between 1 and 9.9) and an exponent (between –9 and 12), or press a preset such as **Virus** or **Earth to Sun**.
- **Predict first:** write the ordinary number you expect, then press Check.
- Use **Exponent + 1** and **Exponent** – **1** to move one rung at a time. The decimal point stays still — it is the digits that slide across the place-value columns.

- The ladder background on the Scratchpad is there for marking where numbers live.

The Practice Companion pathway

Each question has support in this order: **Socratic prompt** → **Hint** → **Check** → **Worked solution**. The worked solution starts locked. It unlocks for a question after you have used the Socratic prompt, used the hint, and had **two tries**. This is not a punishment — it protects your thinking time, and the support path is always visible.

Using support is part of learning. It is not cheating.

When you feel stuck

- Take one small step: write the number you *do* know something about.
- Slide the digits one place at a time and count aloud — the decimal point stays where it is.
- Mark the number on the Scratchpad ladder — which two anchors does it sit between?
- Use a Reconnection Route, then come back. Every route returns you here.

How will I know I am making progress?

- I can turn 8850 into standard form without counting one zero at a time.
- I can say which of two standard-form numbers is bigger by looking at the exponents first.
- I can explain to someone else why 45×10^3 is not in standard form.

Before you finish

Look back at your postcard draft. Which conversion felt smoothest? Which one deserves another visit next time? One idea worth carrying forward is enough.

You can pause. You can return. You do not need to complete everything in one sitting.